MSc Project Sharing and Codex Handover Guide

AI-Driven Weather Forecasting and Crop Recommendation System
for Anuradhapura District

Purpose: A practical guide for safely sharing the GitHub repository with another developer or starting the project in a new Codex session without losing approved project decisions.

Repository	github.com/ICBT-MSC-IT-27-01/final-project.git
Current branch	main
Current sharing status	**SAFE TO SHARE**
Completed development	Phases 1-5 approved/frozen; Phase 5 complete
Next planned phase	Phase 6 - Backend Forecast Integration

Important: owner-specific Codex recovery files and local session metadata are intentionally excluded from the shared repository.

1. What the Recipient Receives

A fresh clone of the repository contains the application source code, project guidance, approved/frozen project context, the ERA5 training dataset, the approved trained LSTM model, scaler, model metadata, evaluation results, training history, sample forecast, and instructions for starting a new Codex session.

- Application source code and automated tests
- AGENTS.md and the project-specific Codex skill
- docs/CODEX_HANDOVER.md
- docs/PROJECT_SHARING_CHECKLIST.md
- docs/anuradhapura_weather_2010_2025_era5.csv
- Approved Phase 5 model artifacts under src/forecasting-service/artifacts/v1_era5_2010_2025/

The repository does not include the owner's local Codex recovery file, AgentSessions directory, or .codex data.

2. How to Share the Project

Share the GitHub repository with the recipient. If the repository is private, grant the recipient the appropriate GitHub access first. Do not send personal Codex session folders, local credentials, or machine-specific recovery files.

Repository to share

https://github.com/ICBT-MSC-IT-27-01/final-project.git

Recommended message to the recipient

Please clone the repository and read START/HANDOVER guidance before changing any code. The project contains approved/frozen MSc decisions that must not be changed without project-owner approval. Do not start a new development phase until you have summarized the current state and received approval.

3. Recipient Setup

The recipient should clone the repository using Git:

```
git clone https://github.com/ICBT-MSC-IT-27-01/final-project.git
cd final-project
```

After cloning, the recipient should first inspect the repository guidance. Dependency installation and runtime setup should follow the project files and existing documentation rather than replacing versions or redesigning the solution.

4. Required Reading Before Any Changes

The recipient or new Codex session should read these in order:

1. AGENTS.md
2. .agents/skills/anuradhapura-ai-development/
3. docs/CODEX_HANDOVER.md
4. docs/PROJECT_SHARING_CHECKLIST.md

These files define the project scope, architecture, frozen decisions, completed phases, model status, unresolved decisions, and the rules a future developer or Codex session must follow.

5. New Codex Session - Startup Instruction

Paste the following instruction into a new Codex session after cloning the repository:

```
Read the following before making any changes:

1. AGENTS.md, if present
2. .agents/skills/anuradhapura-ai-development/
3. docs/CODEX_HANDOVER.md
4. docs/PROJECT_SHARING_CHECKLIST.md

This is an MSc project with approved/frozen development decisions.

Confirm that you understand the project scope, the six supported crops, the architecture, the
completed phases, the trained LSTM model, the distinction between weather forecasting and crop
recommendation, the unresolved decisions, the current project status, and the next planned phase.

Do not modify anything yet. First provide a short project-state summary and wait for approval.
```

Expected behavior

The new Codex session should summarize its understanding first. It should not change source code, schema, model artifacts, scope, or frozen decisions until the project owner approves the next task.

6. Project Scope That Must Remain Fixed

Area	Approved scope
Geography	Anuradhapura District, Sri Lanka
Crops	Paddy, Maize, Green Gram, Cowpea, Groundnut, Chilli
Forecast	7-day rainfall, temperature and humidity forecast
Forecast model	Python LSTM using TensorFlow/Keras
Recommendation	Separate explainable weighted/rule-based suitability engine
Suitability factors	Rainfall, temperature, humidity, soil compatibility

Key design rule: The LSTM predicts future weather. It does not directly predict a crop. The crop suitability engine uses the 7-day forecast together with crop environmental requirements, soil compatibility, and configurable weights/thresholds to calculate explainable suitability results.

7. Approved Architecture and Roles

Technology stack: Angular frontend, ASP.NET Core Web API backend, SQL Server database, and a Python/TensorFlow-Keras forecasting component.

Approved logical flow

```
Angular -> ASP.NET Core Web API -> SQL Server
ASP.NET Core Web API -> Python Forecasting Component -> Saved LSTM Model / Scaler
```

The backend follows a logical three-tier design: Presentation, Business/Application, and Data Access. The practical code structure uses Api, Application, Domain, and Infrastructure. Domain belongs to the business/core side and is not a separate deployment tier.

User roles

Role	Boundary
Public User	Can access core public weather/recommendation functionality without a required database account.
Registered User	Can register/login/logout and access recommendation history/details.
Agricultural Officer	Can review selected recommendations, evidence and validation status; cannot edit suitability configuration.
Administrator	Can manage approved configuration areas, users/roles, and view recommendations/validations.

Normal registration creates only a Registered User. Users must not be allowed to self-assign Agricultural Officer or Administrator roles.

8. Development Status

Phase	Status
1 - Solution / Project Structure	APPROVED / FROZEN
2 - Database Foundation	APPROVED / FROZEN
3 - Authentication / RBAC	APPROVED / FROZEN
4 - Admin Configuration Foundation	APPROVED / FROZEN
5A - ML Pipeline Foundation	APPROVED / FROZEN
5B - Dataset Training and Evaluation	APPROVED / FROZEN
5 - Weather Forecasting Model	COMPLETE / APPROVED / FROZEN
6 - Backend Forecast Integration	NEXT - NOT STARTED

9. Dataset and Approved Forecasting Model

The approved training dataset is **docs/anuradhapura_weather_2010_2025_era5.csv**. It contains ERA5 historical/reanalysis daily weather data accessed through Open-Meteo for a representative Anuradhapura location.

Period	2010-01-01 to 2025-12-31
Records	5,844 daily records
Variables	Temperature, Rainfall, Humidity
Missing values	0
Duplicate dates	0
Continuity	Complete daily continuity

Dataset limitation: ERA5 is reanalysis/modelled weather data, not direct Sri Lanka Department of Meteorology station observations. It must not be described as exact microclimate coverage for every location in Anuradhapura District.

Training configuration

30-day lookback -> 7-day direct multi-step forecast. Input shape: (samples, 30, 3). Output shape: (samples, 7, 3). Training period: 2010-2021; validation: 2022-2023; test: 2024-2025. MinMaxScaler was fitted only on training data. Model: Input(30,3) -> LSTM(64) -> Dropout(0.2) -> Dense(21) -> Reshape(7,3).

Evaluation summary

Metric	Baseline	LSTM
Overall MAE	3.584	3.163
Overall RMSE	8.738	6.729
Rainfall MAE / RMSE	5.493 / 13.916	4.904 / 10.547
Temperature MAE / RMSE	0.793 / 1.040	0.707 / 0.891
Humidity MAE / RMSE	4.465 / 5.859	3.879 / 4.881

The LSTM outperformed the persistence baseline. Rainfall remains the most difficult target. Per-variable metrics should receive greater emphasis than combined overall metrics because the variables use different units.

Approved artifacts in Git

```
src/forecasting-service/artifacts/v1_era5_2010_2025/
- v1.keras
- v1_scaler.joblib
- v1_metadata.json
- v1_evaluation.json
- v1_training_history.json
- v1_sample_forecast.json
```

10. Rules the Recipient Must Follow

1. Treat APPROVED/FROZEN decisions as immutable unless the project owner explicitly approves a change.
2. Do not silently expand the project scope.
3. Do not invent agricultural numeric thresholds or ranges.
4. Do not fabricate or alter model performance results.
5. Do not retrain or replace the approved v1 model without approval.
6. Do not unnecessarily modify completed database, authentication/RBAC, or admin foundations.
7. Work phase-by-phase and stop after each phase with a completion report.
8. Wait for project-owner approval before starting the next phase.

11. Decisions Still Unresolved

The following must not be guessed or invented by a new developer or Codex session. They require explicit project-owner approval:

- Final agricultural numeric thresholds/ranges where not yet approved
- Final soil data source and soil selection method
- Specific production hosting/deployment provider
- Any new feature outside the approved scope
- Any architecture or schema change affecting frozen phases

12. Explicitly Out of Current Scope

Unless approved later: nationwide coverage, IoT soil sensors, automated irrigation, pest/disease prediction, market-price forecasting, native mobile application, full multilingual UI, nationwide alerts, yield prediction, autonomous farming decisions, NPK-based recommendation, organic-matter-based recommendation, land-size recommendation, planting-time recommendation, season-specific weighting, and high-humidity climate-risk indication.

13. Privacy and Local Recovery Files

The shared repository intentionally excludes **docs/CODEX_SESSION.md**, **docs/AgentSessions/**, and **.codex/**. These are owner-specific/local recovery materials and are not the handover mechanism for another user.

A previous Git commit may still contain CODEX_SESSION.md in repository history. No Git history rewriting was performed. The current remote tree excludes the file. The session identifier is not a project credential and should not be used by the recipient.

14. Recipient Acceptance Checklist

Before any development begins, the recipient should be able to confirm all of the following:

[ ]	I cloned the correct repository and am working from the intended branch.
[ ]	I read AGENTS.md and the project-specific Codex skill.
[ ]	I read CODEX_HANDOVER.md and PROJECT_SHARING_CHECKLIST.md.
[ ]	I understand that the crop scope is exactly six crops.
[ ]	I understand that the LSTM forecasts weather and does not directly recommend crops.
[ ]	I understand that Phases 1-5 are approved/frozen and Phase 6 is next.
[ ]	I can see the ERA5 dataset in docs/.
[ ]	I can see the approved v1 model, scaler, metadata and evaluation artifacts.
[ ]	I understand the unresolved decisions and will not invent them.
[ ]	I will summarize the project state and wait for approval before modifying code.

15. What Happens Next

After the recipient/new Codex session has summarized the project correctly and the project owner approves continuation, the next planned development stage is **Phase 6 - Backend Forecast Integration**. The standalone forecasting model is complete; Phase 6 will integrate the approved forecast capability with the ASP.NET Core backend. No Phase 6 implementation should begin merely because the repository has been cloned.

16. Quick Handover Summary

Share the GitHub repository -> recipient clones it -> reads project guidance -> confirms understanding -> waits for approval -> continues from Phase 6 only after approval.

Repository: github.com/ICBT-MSC-IT-27-01/final-project.git

Document status: prepared from the approved/frozen project state and the verified SAFE TO SHARE GitHub handover status.